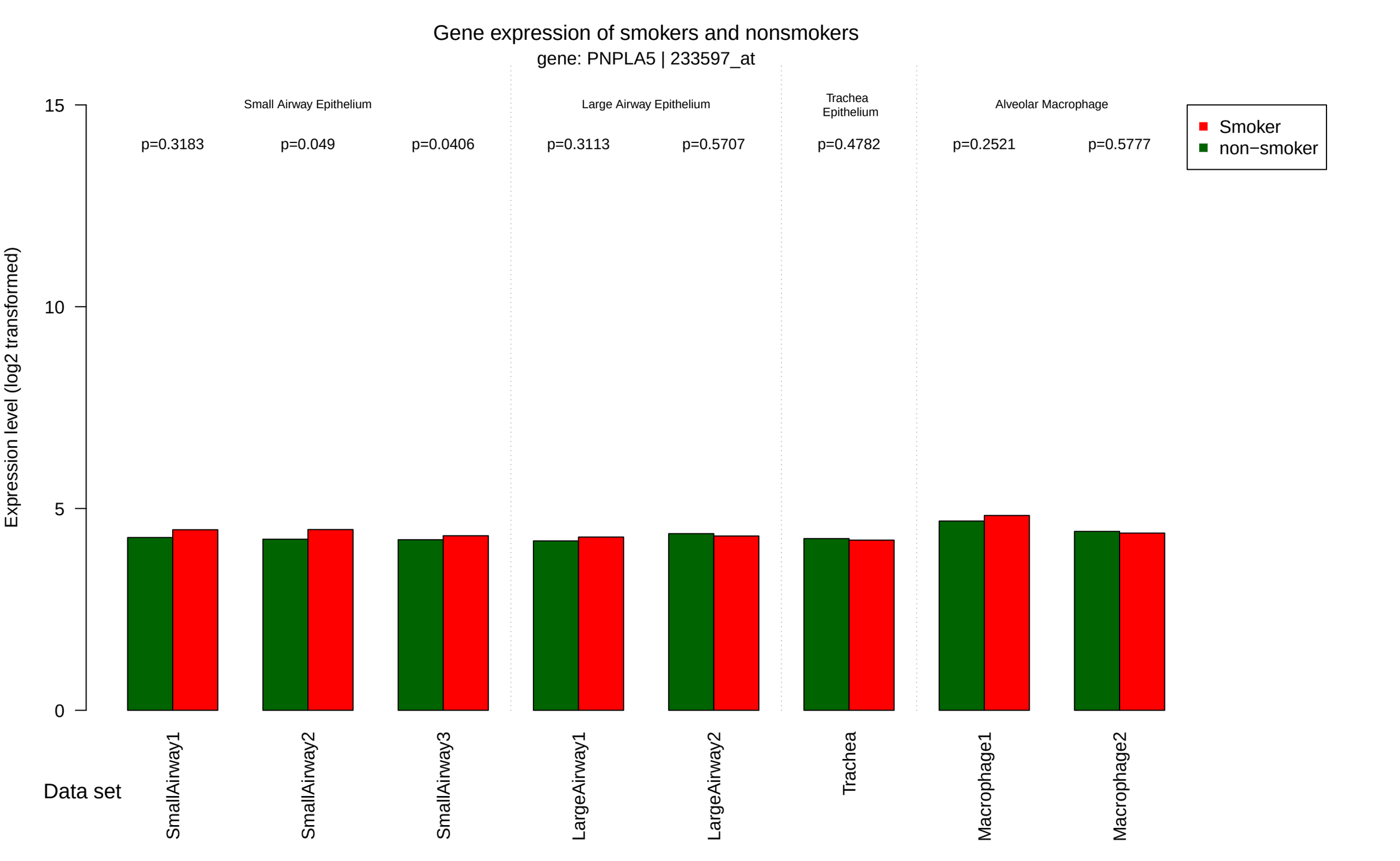
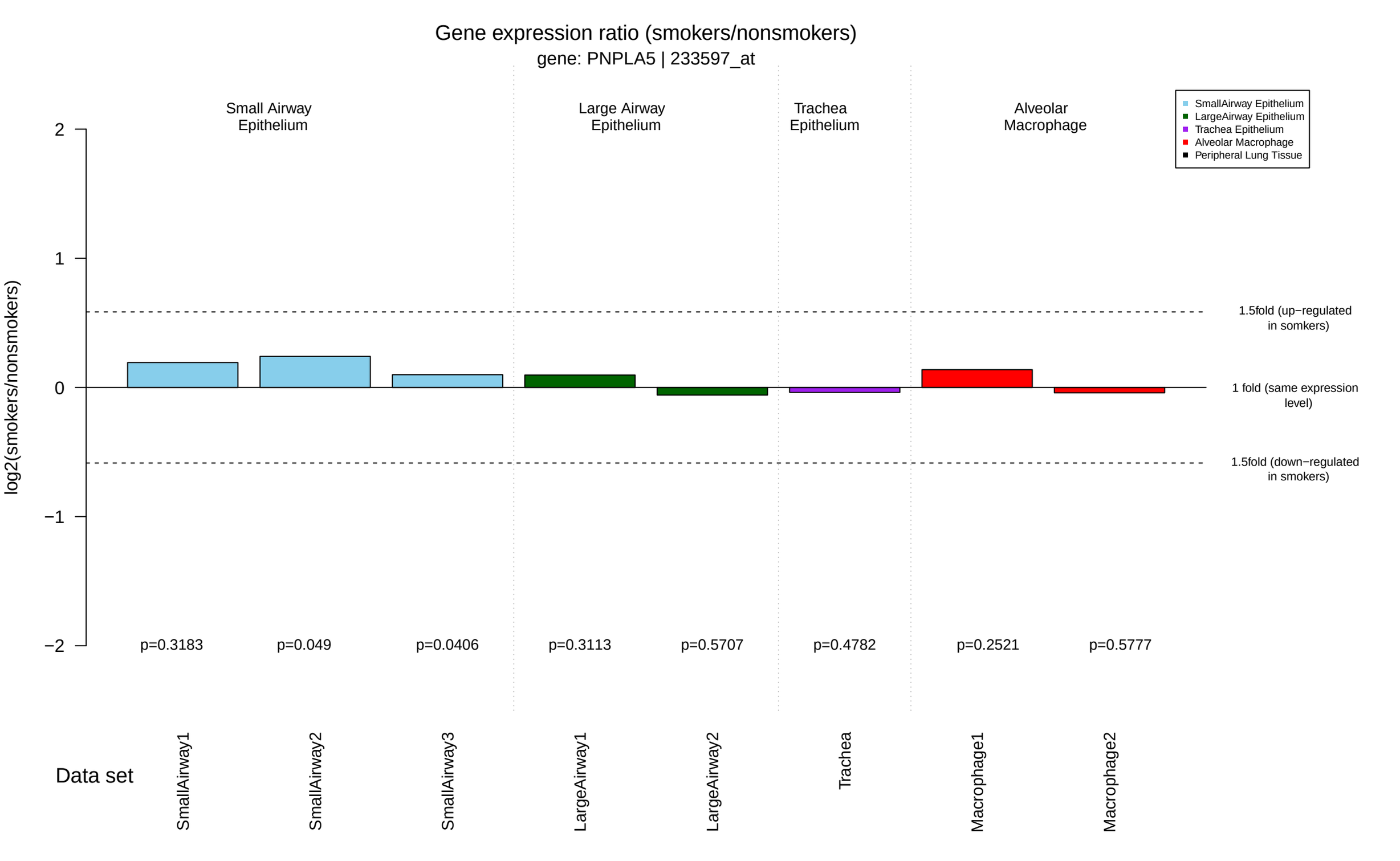
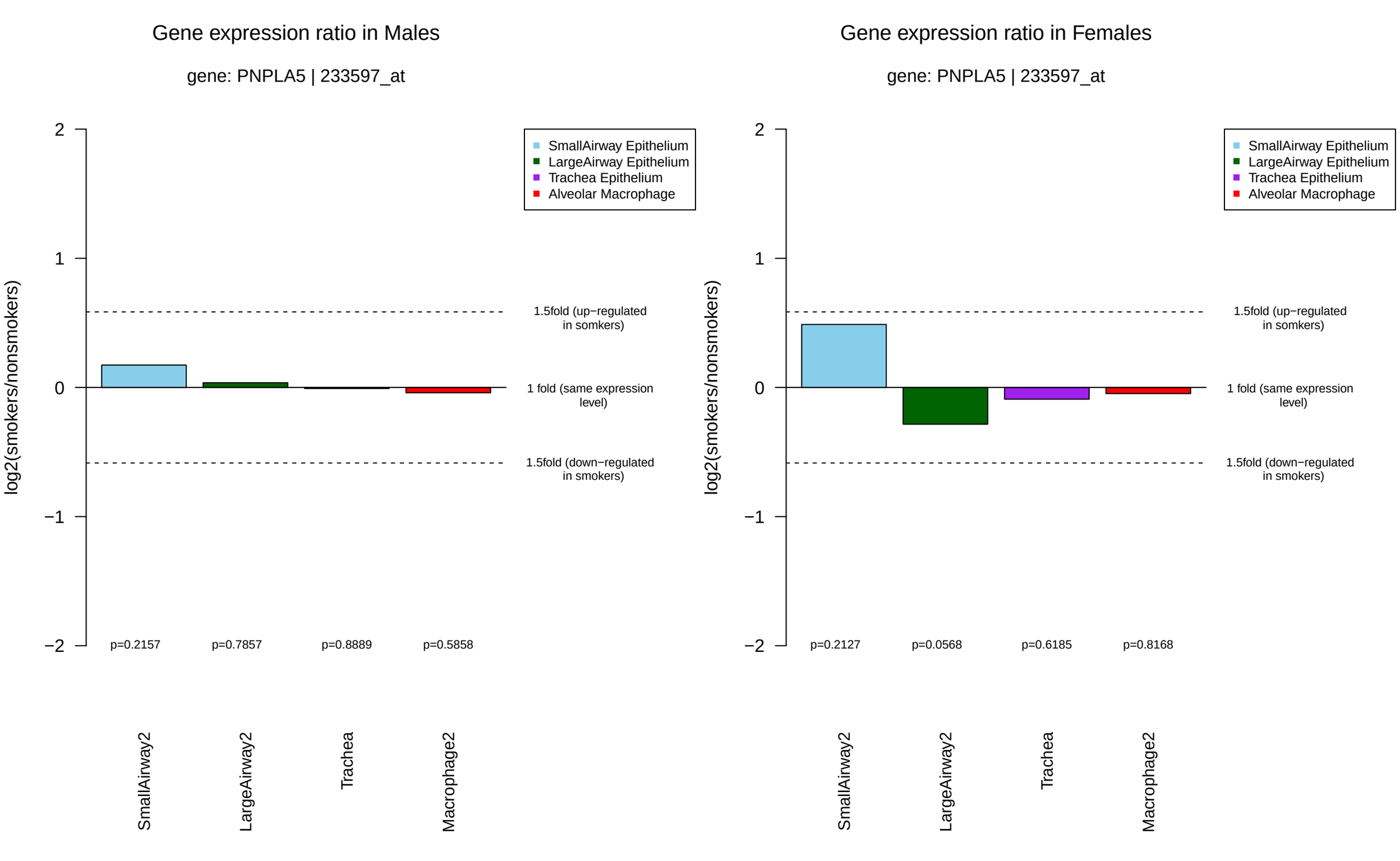
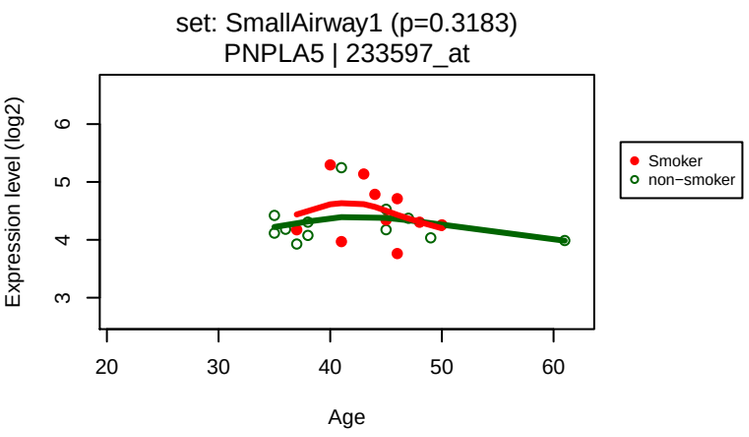
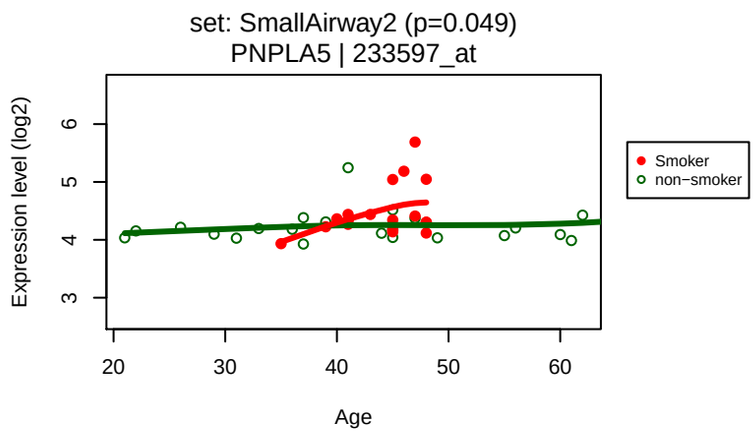
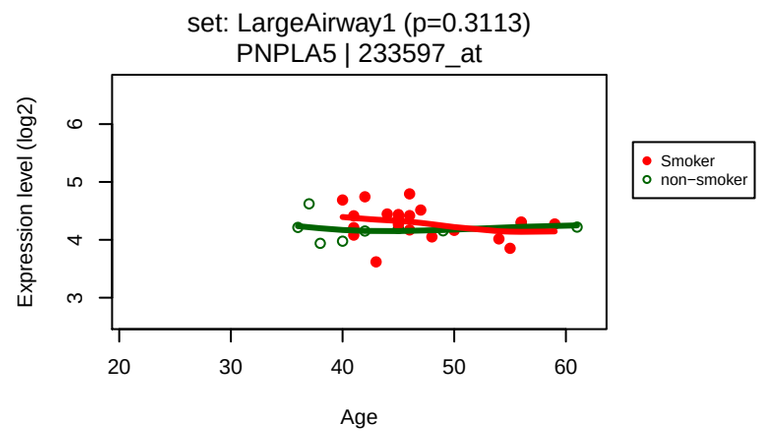

Gene expression of smokers and nonsmokers
gene: PNPLA5 | 233597_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3183
p=0.049
p=0.0406
p=0.3113
p=0.5707
p=0.4782
p=0.2521
p=0.5777
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

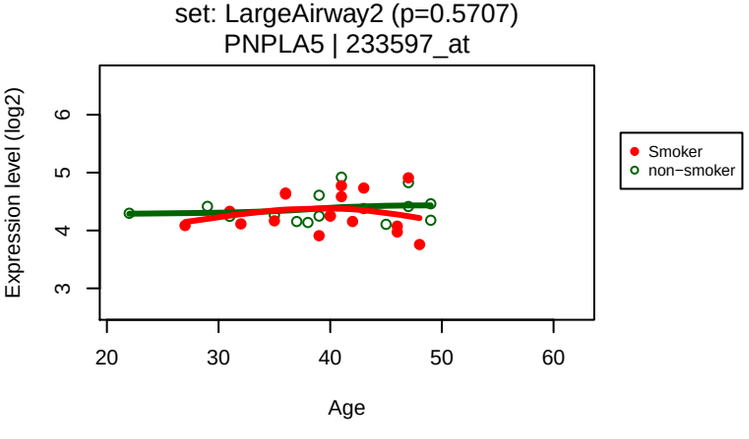

Gene expression ratio (smokers/nonsmokers)
gene: PNPLA5 | 233597_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3183
p=0.049
p=0.0406
p=0.3113
p=0.5707
p=0.4782
p=0.2521
p=0.5777
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

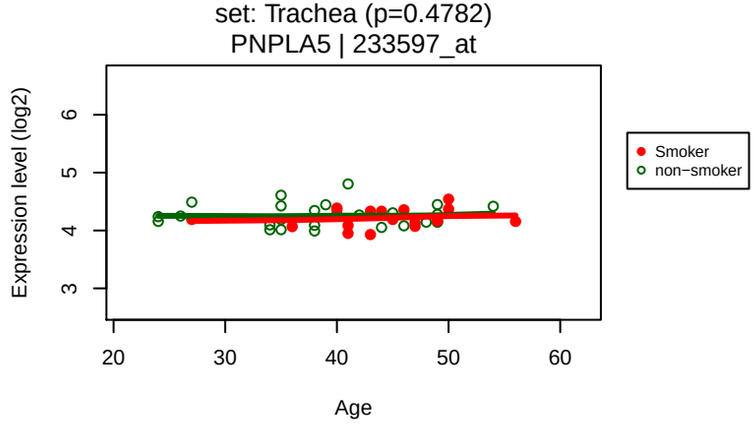

Gene expression ratio in Males
gene: PNPLA5 | 233597_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2157
p=0.7857
p=0.8889
p=0.5858
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PNPLA5 | 233597_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2127
p=0.0568
p=0.6185
p=0.8168
SmallAirway2
LargeAirway2
Trachea
Macrophage2

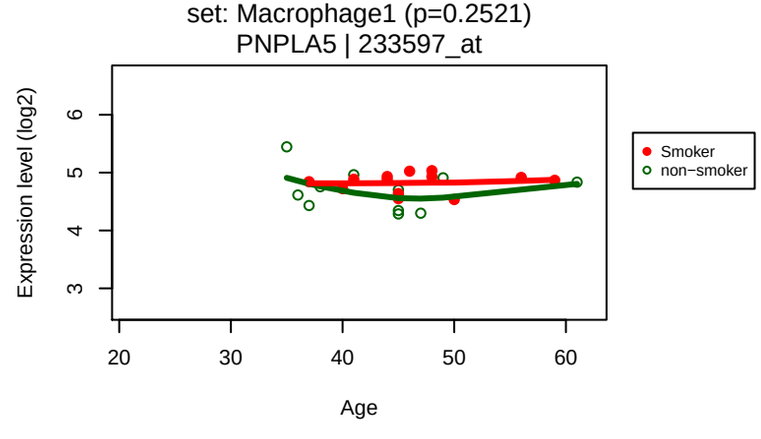

set: SmallAirway1 (p=0.3183)
PNPLA5 | 233597_at
Expression level (log2)
Age
Smoker
non−smoker

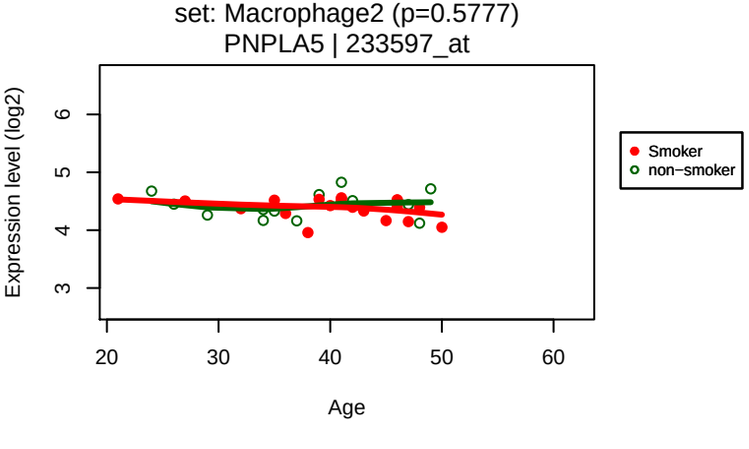

set: SmallAirway2 (p=0.049)
PNPLA5 | 233597_at
Expression level (log2)
Age
Smoker
non−smoker

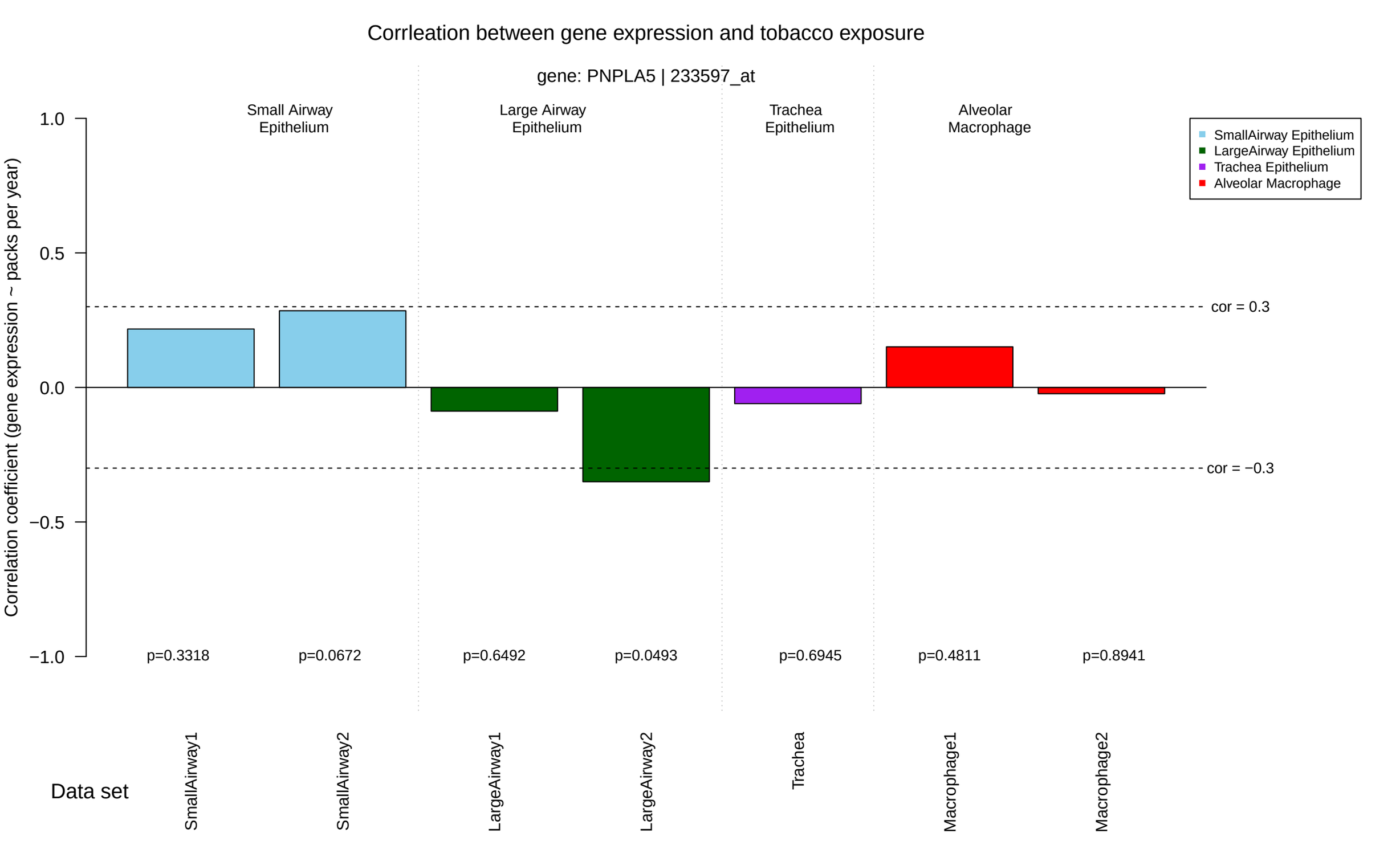

set: LargeAirway1 (p=0.3113)
PNPLA5 | 233597_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5707)
PNPLA5 | 233597_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4782)
PNPLA5 | 233597_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2521)
PNPLA5 | 233597_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5777)
PNPLA5 | 233597_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PNPLA5 | 233597_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3318
p=0.0672
p=0.6492
p=0.0493
p=0.6945
p=0.4811
p=0.8941
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2